



synapseForge

pypi package or version not found python 3.12+ license Apache 2.0

CLI + Template to scaffold and ship AI agent projects (FastAPI + React/Vite/TS)

Description

synapseForge is a PyPI package that provides a CLI to scaffold full-stack AI agent projects from scratch — backend (FastAPI), frontend (React/Vite/TypeScript), branding (logo, .ico, color palette), dependencies, and a self-contained distribution build.

Installation

```
pip install synapseForge
```

Package dependencies (only these two — everything else is project-level):

- `colorthief>=1.0.0` — Automatic color palette extraction from the logo
- `Pillow>=10.0.0` — .ico favicon generation

Quick Start

Create a new project

```
synapseForge init my-project
```

Interactive pipeline:

1. **Input** — Provide logo (absolute path), company name, owner, legal name, repo name, client name, and optional hex colors.
2. **Template** — Extracts the bundled `template.zip` (or downloads from GitHub if not available locally).
3. **Venv** — Creates `./{repo}/` virtual environment.
4. **Deps** — Runs `pip install -r requirements.txt` inside the venv.
5. **Config** — Saves all input as `config/replace.json`.
6. **Logo** — Copies the logo to `frontend/src/assets/logo_empresa.png`.
7. **.ico** — Generates `logo_empresa.ico` via Pillow (16×16 to 256×256).
8. **Colors** — If no colors were entered, extracts up to 3 dominant colors from the logo using colorthief.
9. **Placeholders** — Replaces all XML tags (`<empresa>` , `<cliente>` , `<color_primario>` , etc.) across every project file.

```
cd my-project
# Ready to develop
```

Target directory behavior:

- If it **does not exist** → the command errors out and does nothing.
- If it **exists and is empty** → the template is extracted cleanly.



- If it **exists with files** → the template is extracted on top. Zip files overwrite existing ones. Files already in the directory that are not in the zip **remain** (orphaned).

Build a distributable

```
synapseForge launch ./my-project "MyApp"
```

1. **PyInstaller** — Compiles the backend into a standalone `.exe` .
2. **Frontend** — Runs `npm run build` → generates `frontend/dist/` .
3. **Package** — Bundles `backend.exe`, `frontend/dist/`, `venv`, `launcher`, `.env`, `LICENSE`, `README`, `docs/` into a zip.

Requirements

Tool	Version	Needed for
Python	3.12+	init and launch
Node.js	20+	launch (frontend build)

Generated project structure

```
my-project/
├── backend/
│   ├── main.py           # FastAPI app (CORS, lifespan, routers)
│   ├── routes/           # Chat SSE, Config, Sessions
│   ├── agent/            # Agent framework (Loop, Tools, MCP, Skills, Sessions, Permissions)
│   └── requirements.txt
├── frontend/
│   ├── src/              # React/Vite/TS – Chat, Sidebar, Config
│   └── package.json
├── config/
│   └── replace.json      # Project placeholders
├── docs/
│   ├── tools/            # Tool creation guide
│   └── agents/           # Agent creation guide
├── .env.example
├── .gitignore
├── LICENSE
└── README.md
```

User configuration (`~/ .config/synapseForge/`)

```
~/ .config/synapseForge/
├── skills/              # Installed skills (SKILL.md per skill)
├── tools/               # Custom tools (.py files)
├── agents/              # Agent definitions with permissions (.md)
└── config.json          # MCP server configuration
```



CLI Reference

Command	Description
<code>synapseForge init [dir]</code>	Scaffold a project from the bundled template
<code>synapseForge launch <path> <exe></code>	Build a self-contained distribution
<code>synapseForge --help</code>	Show global help

Tech Stack



License

Apache 2.0

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